



**The University of Azad Jammu and Kashmir,
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Course Name	Data Structure & Algorithm
Submitted to	Engr. Asma Javaid
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Lab No	09
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Tree Data Structure

What is a Tree?

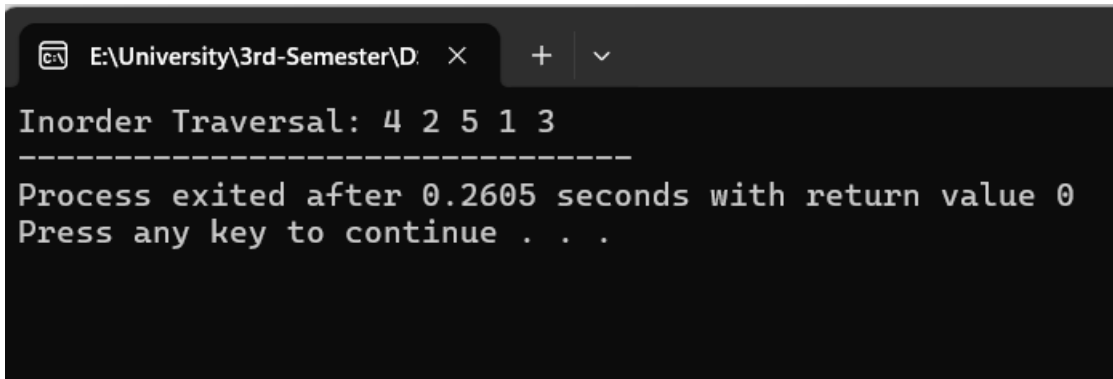
A **Tree** is a non-linear data structure used to store data in hierarchical form. It starts from a **root node**, and every node can have child nodes.

Task No 01:

```
Lab 09.cpp
1  #include <iostream>
2  using namespace std;
3
4  // Structure of a tree node
5  struct Node {
6      int data;           // Data stored in node
7      Node* left;        // Pointer to left child
8      Node* right;       // Pointer to right child
9  };
10
11 // Function to create a new node
12 Node* createNode(int value) {
13     Node* newNode = new Node(); // Allocate memory
14     newNode->data = value;       // Assign data
15     newNode->left = NULL;        // Initialize left child
16     newNode->right = NULL;      // Initialize right child
17     return newNode;
18 }
19
20 // Inorder Traversal: Left -> Root -> Right
21 void inorder(Node* root) {
22     if (root == NULL)
23         return;
24
25     inorder(root->left);
26     cout << root->data << " ";
27     inorder(root->right);
28 }
29
30 int main() {
31     // Creating the tree
32     Node* root = createNode(1);
```

```
33     root->left = createNode(2);
34     root->right = createNode(3);
35     root->left->left = createNode(4);
36     root->left->right = createNode(5);
37
38     cout << "Inorder Traversal: ";
39     inorder(root);
40
41     return 0;
42 }
43
```

Output:



```
E:\University\3rd-Semester\D... × + v
Inorder Traversal: 4 2 5 1 3
-----
Process exited after 0.2605 seconds with return value 0
Press any key to continue . . .
```

Tree Traversals:

1. **Inorder Traversal:** Left → Root → Right
2. **Preorder Traversal:** Root → Left → Right
3. **Postorder Traversal:** Left → Right → Root